Lecture 14: Geometric Buckling and the Reflected Core

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Reading Assignment

Lamarsh & Baratta (4th Edition):

- **Chapter 6:** Nuclear Reactor Theory.
  - Sections 6.2 – 6.4: Geometric Buckling and Criticality.

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1 The Chemical Engineering Analogy: Autocatalysis

Before we solve the reactor equation, let's recognize that this is not a new problem. It is a standard Chemical Engineering problem in disguise.

Consider an **Autocatalytic Reaction** occurring in a gel slab, where a species A catalyzes its own production:

$$A + \ldots \xrightarrow{k} 2A + \ldots$$

The rate of production is proportional to the concentration C_A. The steady-state diffusion equation for species A is:

$$\underbrace{\mathcal{D}\nabla^2 C_A}_{\text{Diffusion}} + \underbrace{kC_A}_{\text{Production}} = 0$$

Rearranging:

$$\nabla^2 C_A + \frac{k}{\mathcal{D}} C_A = 0$$

Now look at our **Reactor Equation** from Lecture 13:

$$\nabla^2 \phi + B_m^2 \phi = 0$$

They are identical. A nuclear reactor is simply a diffusion-limited autocatalytic reactor.

- If the slab is too thin, the reactant A (or neutrons) diffuses out the sides faster than it is created. The concentration drops to zero.
- If the slab is thick enough, production exceeds leakage, and the concentration explodes (or sustains itself).

2 The Infinite Slab Reactor

Let's solve this for the simplest geometry: an infinite slab of thickness L (from $x = -L/2$ to $x = +L/2$). The reactor equation in 1D is:

$$\frac{d^2\phi}{dx^2} + B_m^2\phi = 0 \tag{1}$$

2.1 Boundary Conditions (The Vacuum Approximation)

Neutrons that leave the slab do not come back.

- At the center ($x = 0$), the flux must be symmetric: $\frac{d\phi}{dx} = 0$.
- At the edge ($x = \pm L/2$), the flux must go to zero.
- Note: In the vacuum approximation, we impose zero flux at the boundary; more precisely, the flux extrapolates to zero just outside the physical surface.

2.2 The Solution (Sturm-Liouville Problem)

This is a classic eigenvalue problem. The general solution is:

$$\phi(x) = A\cos(B_m x) + C\sin(B_m x)$$

Applying symmetry ($d\phi/dx = 0$ at $x = 0$) kills the sine term ($C = 0$).

$$\phi(x) = A\cos(B_m x)$$

Applying the boundary condition $\phi(L/2) = 0$:

$$\cos\left(B_m\frac{L}{2}\right) = 0$$

As with all Sturm-Liouville problems, there is a trivial solution: the flux is zero everywhere! In order for a non-trivial solution to exist, the buckling B_m must take on particular values (the eigenvalues of the problem). For this boundary condition, $B_m x$ evaluated at the boundary must be an odd multiple of $\pi/2$:

$$B_m\frac{L}{2} = \frac{n\pi}{2} \quad \text{for } n = 1, 3, 5\ldots$$

2.3 The Fundamental Mode

While mathematical solutions exist for $n = 3, 5, \ldots$, these represent higher harmonic modes.

- For a linear problem such as this, any initial condition (e.g., flux or concentration profile) can be expanded as a wieghted sum of these eigenfunctions.
- **Time Dependence:** The eigenvalues associated with each eigenfunction tells you how fast that particular term decays away - and very quickly your profile is dominated by the lead eigenvalue and eigenfunction - the fundamental mode.

Therefore, the only stable, steady-state solution is the **Fundamental Mode** ($n = 1$):

$$B_m = \frac{\pi}{L}$$

Squaring this gives us the **Geometric Buckling (B_g^2)** for a slab:

$$B_g^2 = \left(\frac{\pi}{L}\right)^2 \tag{2}$$

3 Criticality: The Balance

For the reactor to be at steady state with a non-trivial solution, the **Material Buckling** (what the fuel *can* do) must equal the **Geometric Buckling** (what the size *requires*):

$$B_m^2 = B_g^2 \tag{3}$$

$$\frac{k_\infty - 1}{L_{diff}^2} = \left(\frac{\pi}{L_{phys}}\right)^2$$

This equation allows you to calculate the Critical Width (L_{phys}) required for a given fuel mixture. Widths less than this would not sustain a chain reaction, widths greater than this would cause the neutron flux to increase exponentially (a reactor works only because of negative response coefficients: as the flux (and temperature) go up, the material buckling goes down to match B_g).

4 The Reflected Slab (The "Biot Number" Analogy)

Real reactors are surrounded by a **Reflector** (water, graphite, or beryllium) that bounces neutrons back into the core. How do we model this without solving two coupled differential equations? We change the boundary condition.

4.1 The Robin Boundary Condition

Instead of the flux dropping to zero at the edge ($\phi = 0$), the reflector imposes a relationship between the flux and its gradient (current).

- By Fick's Law: $J_{out} = -D\frac{d\phi}{dx}$.
- The reflector returns a fraction of this current.

This creates a boundary condition mathematically identical to **Newton's Law of Cooling** in heat transfer:

$$-D\frac{d\phi}{dx}\bigg|_{surface} = h_{eff}(\phi_{surface} - 0) \tag{4}$$

Rearranging to look like a Biot Number condition:

$$\frac{1}{\phi}\frac{d\phi}{dx} = -\frac{h_{eff}}{D} = -\frac{1}{\lambda_{reflector}}$$

4.2 The Result: Reflector Savings

If we solve the diffusion equation with this "leaky" boundary condition, the cosine shape doesn't have to hit zero at the wall. It can stay finite.

The effect is that the reactor "feels" bigger than it physically is.

- **Vacuum:** Flux hits zero at $L/2$.
- **Reflected:** Flux would hit zero at some extrapolated distance $L/2 + \delta$.

We call δ the **Reflector Savings**. It allows us to build the physical core smaller ($L_{new} = L_{old} - 2\delta$) while maintaining the same criticality.

5 Geometric Buckling for Other Shapes

We can apply the same method (Helmholtz equation $\nabla^2\phi + B^2\phi = 0$) to other geometries. Rather than deriving the Bessel functions (Cylinder) or Spherical Bessel functions (Sphere) in class, we summarize the results. For a reactor to be critical, B_m^2 must equal B_g^2.

Geometry	**Dimensions**	**Flux Profile** $\phi(r)$	**Geometric Buckling** B_g^2
Infinite Slab	Thickness L	$\cos(\frac{\pi x}{L})$	$\left(\frac{\pi}{L}\right)^2$
Sphere	Radius R	$\frac{1}{r}\sin(\frac{\pi r}{R})$	$\left(\frac{\pi}{R}\right)^2$
Finite Cylinder	Radius R, Height H	$J_0(\frac{2.405r}{R})\cos(\frac{\pi z}{H})$	$\left(\frac{2.405}{R}\right)^2 + \left(\frac{\pi}{H}\right)^2$

Table 1: Geometric Buckling for standard reactor shapes.

Note that the solution to this equation for very many geometries (and including finite Biot numbers) can be found in the heat transfer literature: Carslaw, H. S., & Jaeger, J. C. (1959). Conduction of Heat in Solids (2nd ed.). Oxford: Clarendon Press.

6 Case Study: The Cecil Kelley Accident (1958)

To understand why Geometric Buckling matters, consider the tragic accident of Cecil Kelley at Los Alamos National Laboratory.

6.1 The Setup

Kelley was a chemical operator working with a large mixing tank (1000 liters). The tank contained two immiscible liquids:

1. A heavy aqueous phase (acidic water).
2. A light organic phase (containing dissolved Plutonium) floating on top.

The Plutonium concentration in the organic layer was high ($>$ 3 kg total), but the layer was relatively thin.

- **Geometry:** Slab-like (High Surface Area / Volume ratio).
- **Result:** High leakage. B_g^2 (slab) was large. $B_g^2 > B_m^2$. The system was **Sub-critical**.

6.2 The Trigger: Turning on the Mixer

When Kelley turned on the stirrer to mix the phases, the fluid dynamics changed the geometry of the plutonium layer.

- The vortex created by the mixer drew the organic layer down into the center of the tank.
- The thin "slab" became a central "blob" (roughly spherical or cylindrical).

6.3 The Consequence

- **Geometry Change:** The "blob" had a much lower Surface Area / Volume ratio than the slab.
- **Result:** Neutron leakage dropped dramatically. B_g^2 (sphere) is much smaller than B_g^2 (slab) for the same volume.
- **Criticality:** Suddenly, $B_g^2 < B_m^2$.

The solution went Prompt Critical. A "Blue Flash" (Cherenkov radiation) was observed. Kelley received a lethal dose of neutrons and gamma rays ($\sim 12,000$ Rads) and died 35 hours later. **Lesson:** Criticality is not just about Mass. It is about **Shape**. A safe slab can become a lethal sphere simply by stirring it.

7 Optical Phenomena in Criticality Excursions

The visual observation of a "blue flash" during a criticality accident is a distinct signature of a prompt critical excursion. This luminosity arises from two distinct physical mechanisms: external ionization of the atmosphere and internal Cherenkov radiation generated within the ocular media of the observer.

7.1 Mechanism 1: Ionized Air Glow (External)

The intense flux of gamma rays and neutrons released during the excursion ionizes the surrounding air. The characteristic blue glow results from the de-excitation of nitrogen and oxygen molecules.

- **Process:** High-energy particles eject electrons from atmospheric molecules (N_2, O_2).
- **Emission:** As excited N_2^+ ions and excited N_2 molecules relax to their ground states, they emit photons primarily in the blue and violet spectral regions (391.4 nm and 427.8 nm bands).
- **Observation:** This creates a volumetric blue halo around the source, visible to anyone with a line of sight.

7.2 Mechanism 2: Intraocular Cherenkov Radiation (Internal)

The second mechanism explains reports where victims perceive a flash even if their eyes are closed, or describe the light as originating "inside" their head. This is caused by relativistic charged particles passing directly through the vitreous humor of the eye.

Cherenkov radiation occurs when a charged particle travels through a dielectric medium with velocity v exceeding the speed of light in that medium (c_n):

$$v > \frac{c}{n(\lambda)} \tag{5}$$

where c is the speed of light in vacuum and $n(\lambda)$ is the refractive index of the medium.

For the human eye, the vitreous humor is composed largely of water with a refractive index $n \approx 1.33$. The threshold velocity for Cherenkov emission is therefore:

$$v_{thresh} \approx \frac{c}{1.33} \approx 0.75c \tag{6}$$

During a criticality accident, secondary electrons produced by gamma interactions (Compton scattering) often exceed this velocity threshold, which corresponds to approximately $0.26 MeV$. As these electrons traverse the eyeball, they generate a photonic shockwave in the blue/UV spectrum (intensity $I \propto \frac{1}{\lambda^2}$), which is directly detected by the retina.

7.3 Validation: The Apollo Light Flashes

The biological reality of intraocular Cherenkov radiation was confirmed by the Apollo lunar missions.

- **Observation:** Apollo 11 astronauts (Aldrin, Armstrong, and Collins) reported seeing bright flashes and streaks of light while in the darkened command module, even with their eyes closed.
- **Experiment:** The Apollo Light Flash Moving Emulsion Detector (ALFMED) flown on Apollo 16 and 17 confirmed that these flashes coincided with cosmic rays (high-energy protons and heavy ions) traversing the astronauts' eyes [1, 2].
- **Conclusion:** The flashes are believed to result from high-energy particles traversing the eye, likely through a combination of Cherenkov radiation and direct retinal stimulation, confirming that high-energy radiation can be perceived visually without external light sources.

References

[1] Pinsky, L. S., Osborne, W. Z., Bailey, J. V., Benson, R. E., & Thompson, L. F. (1974). *Light Flashes Observed by Astronauts on Apollo 11 through Apollo 17*. Science, 183(4128), 957-959.

[2] P. J. McNulty et al. ,*Visual Sensations Induced by Relativistic Nitrogen Nuclei*.Science178,160-162(1972).DOI:10.1126/science.178.4057.160

8 The Criticality "U-Curve": Mass vs. Moderation

Before analyzing accident scenarios, we must understand the relationship between Critical Mass and Moderation. This is determined by two competing physical effects:

1. **Thermalization (Decreases Critical Mass):** Adding water slows neutrons down. For ^{239}Pu, the fission cross-section rises from ~ 2 barns (Fast) to ~ 750 barns (Thermal). This makes the fuel much more efficient.
2. **Parasitic Absorption (Increases Critical Mass):** Hydrogen is a neutron poison ($\sigma_a \approx 0.33$ b). If there is too much water (high dilution), the hydrogen atoms outnumber the fuel atoms by so much that they absorb the neutrons before they can find a fuel nucleus.

The result is a "U-shaped" curve (or "bucket").

8.1 The Data (Ref: LA-10860-MS)

The minimum critical mass does *not* occur at maximum density. It occurs at an "Optimum Moderation" ratio.

State	H / Pu Ratio	Critical Mass (^{239}Pu)
Pure Metal (Fast)	0	$\approx$ **10** kg
Undermoderated Sludge	10 – 100	$\approx 2-5$ kg
Optimum Solution (Thermal)	~ 900	**0.51** kg
Infinite Dilution	> 3600	∞ (Always Subcritical)

Table 2: Critical Mass of Plutonium-Water mixtures. Note that the "Optimum" state requires $\sim 20\times$ less fuel than the pure metal.

9 Case Study 2: The Hanford Pulse Jet Mixers

A modern example of this "Moderation Trap" arose during the design of the Hanford Waste Treatment Plant (WTP).

9.1 The Engineering Problem

The waste stored in massive tanks during processing must be mixed with chemicals for treatment. The WTP design uses very large mixing vessels (10m tall) where the chemicals are combined with the waste using **Pulse Jet Mixers (PJMs)**. These are designed to mix the vessels by "sucking fluid in" to a large diameter tube and then "spitting it back out," ideally mixing things up without any moving parts (just using air pressure). Engineers were initially concerned about PuO_2 particles settling in the "elbows" of the piping, fearing a pile-up.

9.2 The "Cloud" Hazard

A criticality safety review identified the inverse risk:

1. **The Settled State (Safe):** Suppose there is 2 kg of PuO_2 settled on the bottom. It is dense and undermoderated. Since the critical mass for this "sludge" is high ($>$ 2 kg), the system is **Subcritical**.
2. **The Pulse (Unsafe):** When the mixer activates, it sucks this sludge up and disperses it with water inside the pulse jet volume.
3. **The Transition:** The dense sludge expands into a dilute "cloud."
   - The system moves down the U-curve from "Undermoderated" to "Optimum."
   - The critical mass drops to **0.5** kg.
4. **Result:** The 2 kg of Plutonium, which was safe on the floor, now could be as many as **4 Critical Masses** simply because it was mixed with water. Such a configuration could, in principle, exceed criticality during the mixing transient.

9.3 Epilogue: The Status Today

The concerns regarding these Pulse Jet Mixers were significant enough that the Department of Energy halted construction on the Pretreatment Facility in 2012.

- **The Validation:** In 2017, full-scale testing of a prototype "Standard High Solids Vessel" was completed to prove that revised control strategies could prevent the accumulation of fissile solids.

- **The Pivot:** Despite this validation, the facility remains unfinished. The site has pivoted to a "Direct-Feed" strategy (DFLAW) that bypasses these mixing vessels entirely to allow low-activity waste processing to begin (slated for $\sim$ 2025), deferring the high-level plutonium sludge processing to the 2030s.

Reference:

PNNL-18098: Pulse Jet Mixing Tests With Noncohesive Solids